

CAMLS UWB/IMU Observability Analysis

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1 System Background

The system in question is a three-drone cable-suspended load carrying formation. It is assumed that:

1. The cable direction is always measurable
2. The cable is always taut
3. The cable is modeled as a mass-spring-damper along the cable axis

In this case the set of drones is $\mathcal{D} = \{1, 2, 3\}$. A diagram of the system with annotated measurements and states can be seen below:

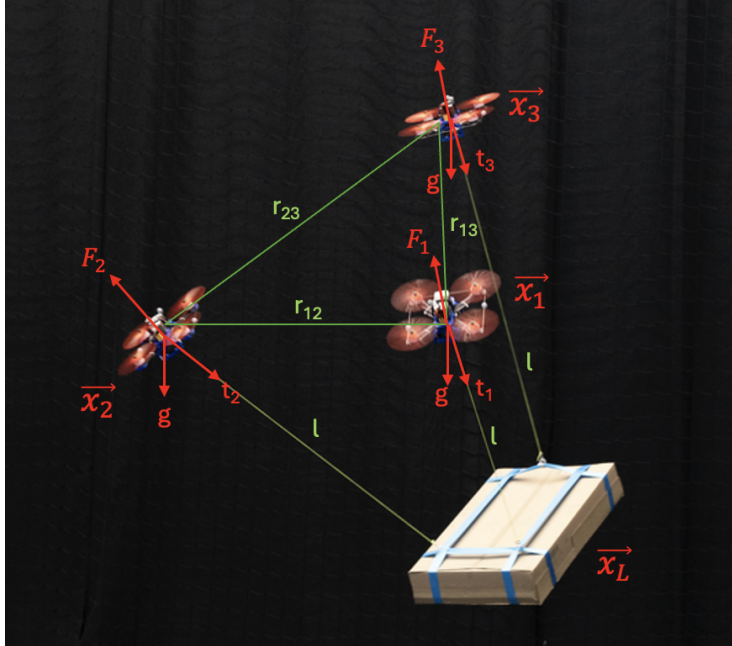


Figure 1: Diagram of System with States and Sensor Measurements

2 Sensors

All drones in the system are equipped with IMUs providing specific force and angular velocity measurements, defined respectively as ${}^b\hat{f}_i$ and ${}^b\hat{w}_i$. In addition, the drones have UWB peer-to-peer ranging, which obtains all inter-agent distances $r_{ij} = \|p_i - p_j\| \forall i, j \in \mathcal{D}$. Finally, drone 1 is assumed to have its full MoCap-derived states ${}^w\hat{p}_1, {}^w\hat{q}_1$.

3 States

As states, we consider the full 6DOF state of the payload and each drone (in set of drones $\mathcal{D}$):

$$\vec{x}_L := {}^w [\vec{p}_L^\top \quad \vec{v}_L^\top \quad \vec{\Theta}_L^\top \quad \vec{\omega}_L^\top]^\top, \quad (1)$$

$$\vec{x}_i := {}^w [\vec{p}_i^\top \quad \vec{v}_i^\top \quad \vec{\Theta}_i^\top \quad \vec{\omega}_i^\top]^\top, \quad \forall i \in \mathcal{D} \quad (2)$$

Combining these, the full system state is:

$$\vec{x} := {}^w [\vec{x}_L^\top \quad \vec{x}_1^\top \quad \dots \quad \vec{x}_N^\top]^\top \quad (3)$$

This means that there are 48 total states (4 bodies with linear and angular positions and velocities). The cable state estimate can be calculated from these, as is done in the current `ekf_cable_state` ros package.

3.1 Notes on rotational representation

There is an important choice when formulating an EKF with rotational states. Rotation Matrices are easy to use in dynamic models, but they exist in $SO(3)$ instead of vector space, so they cannot be used directly in the EKF predict and update steps. With quaternions there is the problem of redundancy (carrying four variables instead of 3) and the EKF update breaks the unit normalization, which then requires a distorting re-normalization after the update step. The initial observability analysis attempted to formulate the C and A matrices with quaternions but found a rank deficit of 4, corresponding to the redundant dimensions of the quaternion. Euler angles are a minimal representation, but they carry the problems of gimbal lock. For now, I will continue with Euler Angles to demonstrate system observability, but in the future I am planning on creating another file formulating the problem as an error-state EKF with quaternions as the state instead of euler angles. For applying rotations I will use rotation matrix notation, which can be obtained from the euler angles.

4 Measurement

The system measurements are as follows:

1. Drone 1 MoCap Position: $\hat{p}_1 = p_1$
2. Drone 1 MoCap Orientation $\hat{\Theta}_1 = \Theta_1$
3. Inter-Drone Ranges: $\hat{r}_{12}, \hat{r}_{13}, \hat{r}_{23}$
4. Body-Fixed Cable Directions: $\hat{s}_i = R_i^T q_i$
5. Accelerometer Specific Force: ${}^b f_i = R_i^T (\dot{v}_i - \vec{g}) = \frac{(\vec{F}_i + \tau_i \vec{s}_i)}{m_i}$
6. Gyro Rotation Rates: ${}^b \hat{w}_i$

5 Inputs

The only control inputs of the system are from the drone rotor forces. This is simplified in this model into the overall force ${}^b F_i$ and ${}^b M_i$. With a well-defined motor and drone dynamics model, these can be calculated from motor RPM.

6 Geometry, Kinematics, and Dynamics

Please note that in this section all frame rotations are expressed as the rotation matrices which can be obtained with the orientation measurement.

6.1 Geometric Constraints

The load cable attachment point constraints can be expressed as:

$${}^w c_i = {}^w p_L + R_L \rho_i \quad (4)$$

where ρ_i is the cable attachment point in the load frame. The drone cable attachment point can be expressed as:

$${}^w h_i = {}^w p_i + R_i d_i \quad (5)$$

where d_i is the cable attachment point in the drone frame. Define the cable direction as q_i where:

$$e_i = c_i - h_i, \quad l_i = \|e_i\|, \quad q_i = \frac{e_i}{l_i} \quad (6)$$

Thus, the cable tension from the mass-spring-damper equation can be expressed for each cable as:

$$\tau_i = k(l_i - l_{rest}) + c(\dot{c}_i - \dot{h}_i) \cdot s_i \quad (7)$$

Note that the s_i points from the drone to the cable.

6.2 Dynamics Equations

Now that the mass-spring-damper forces have been derived, the overall rigid-body dynamics of all four bodies in the system can be found. For the load, the dynamics can be expressed as:

$$\dot{p}_L = v_L, \quad (8)$$

$$m_L \dot{v}_L = m_L \vec{g} - \sum_i \tau_i \vec{q}_i \quad (9)$$

$$\dot{\Theta}_L = T(\Theta)^{-1} {}^b w_i \quad (10)$$

$$J_L \dot{w}_L = \sum_i \tau_i (\rho_i \times R_L^T q_i) - w_L \times J_L w_L \quad (11)$$

For each drone i , the dynamics can be expressed as:

$$\dot{p}_i = v_i, \quad (12)$$

$$m_i \dot{v}_i = R_i({}^b \vec{F}_i) + \tau_i \vec{q}_i + m_i \vec{g} \quad (13)$$

$$\dot{\Theta}_i = T(\Theta_i)^{-1} {}^b w_i \quad (14)$$

$$J_i \dot{w}_i = M_i + \tau_i (d_i \times R_i^T q_i) - w_i \times J_i w_i \quad (15)$$

Because we are using euler angles, we have to propagate the body-fixed angular rates into the euler frame using $T(\Theta)$:

$$T(\Theta) = [e_x \mid R_x(\phi)^T e_y \mid (R_y(\theta) R_x(\phi))^T e_z] = \begin{bmatrix} 1 & 0 & -\sin \theta \\ 0 & \cos \phi & \sin \phi \cos \theta \\ 0 & -\sin \phi & \cos \phi \cos \theta \end{bmatrix} \quad (16)$$

This exposes the limitation of the Euler angle method (at $\phi = \pm 90^\circ$ the determinant of T is 0).

In addition, to aid in the construction of the Jacobians, the cable endpoint dynamics are:

$$\dot{c}_i = v_L + R_L(\omega_L \times \rho_i), \quad \dot{h}_i = v_i + R_i(\omega_i \times d_i), \quad \dot{e}_i = \dot{c}_i - \dot{h}_i. \quad (17)$$

7 Equilibrium

To solve for the equilibrium conditions, we must have a condition where the payload states are static, i.e. $\dot{p}_L = \dot{v}_L = \dot{\Theta}_L = \dot{w}_L = 0$. With the auxiliary variable $w_i = \tau_i q_i$ to simplify the equations, the cable force vectors balance out to:

$$\boxed{\sum_{i=1}^3 w_i = m_L \vec{g}, \quad \sum_{i=1}^3 \rho_i \times w_i = 0.} \quad (18)$$

These are six scalar equations in the nine unknowns $w_1, w_2, w_3 \in \mathbb{R}^3$. Written as a linear system $\mathcal{A}w = b$ with $w \in \mathbb{R}^9$,

$$\mathcal{A} = \begin{bmatrix} I & I & I \\ \rho_1^\times & \rho_2^\times & \rho_3^\times \end{bmatrix} \in \mathbb{R}^{6 \times 9}, \quad b = \begin{bmatrix} m_L \vec{g} \\ 0 \end{bmatrix}. \quad (19)$$

Its solution set is a three-parameter affine family:

$$w = w_0 + N\lambda, \quad \lambda \in \mathbb{R}^3, \quad (20)$$

with w_0 any particular solution and $N \in \mathbb{R}^{9 \times 3}$ a basis for $\ker \mathcal{A}$. The three free parameters are the internal-tension (splay) freedom of a three-cable suspension.

The symmetric family. If instead the geometry is prescribed — all three cables splayed outward by an angle α from vertical, with three-fold symmetry (120deg offset from one another) then (18) is satisfied by inspection and only the magnitudes are needed. Each cable carries a vertical component $\tau \cos \alpha$, so

$$\tau_1 = \tau_2 = \tau_3 = \frac{m_L g}{3 \cos \alpha}, \quad (21)$$

the horizontal components cancel by symmetry, and the moments cancel because each $\rho_i \times w_i$ is tangential and the three are at 120°. If the w is set, then putting the payload at the origin with some R_L defines the geometry of the system.

8 Drone trim

Each drone's balance equations are solved for its inputs(trim). Balancing the force gives

$$R_i F_i = m_i g \hat{z} - \tau_i q_i. \quad (22)$$

For a quadrotor the thrust acts along body z , $F_i = T_i \hat{z}_b$, so this single equation determines both the thrust magnitude and the drone's tilt:

$$\boxed{T_i = \|m_i g \hat{z} - \tau_i q_i\|, \quad R_i \hat{z}_b = \frac{m_i g \hat{z} - \tau_i q_i}{\|\cdot\|}. \quad (23)}$$

The second expression **fixes each drone's roll and pitch** — the drone tilts into its cable — and leaves its **yaw** ψ_i **free**. Moment balance then fixes the torque input,

$$M_i = -\tau_i (d_i \times R_i^T q_i), \quad (24)$$

which vanishes when the hook is at the drone's centre of mass.

9 Linearization

9.1 The chain rule through the cable

Each cable contributes six Jacobians; every row block of A and C is then a short algebraic combination of them. Computing them once per cable per time step, and reusing them, is both the clearest way to present the derivation and the efficient way to implement it.

9.1.1 Rules to Help Derive Cable Derivatives

Rule 1 Differentiating $R(\Theta)$ and using $\partial R_a(\alpha)/\partial\alpha = R_a(\alpha)e_a^\times$ for a single-axis rotation gives

$$\frac{\partial R}{\partial \Theta_k} = R t_k^\times, \quad T(\Theta) = [t_1 \mid t_2 \mid t_3] \quad (25)$$

Rule 2: Differentiation Helpers for Rule 1 Everything in Section 9.1 follows from (25) plus the product rule. For a *constant body-frame* vector a , using $t^\times a = -a^\times t$ and collecting the three columns,

$$\frac{\partial(Ra)}{\partial \Theta_k} = R t_k^\times a = -R a^\times t_k \implies \boxed{D(Ra) = [-\bar{R} a^\times \bar{T}]_\Theta} \quad (26)$$

For a *world-frame* vector b which may itself depend on the state, using $-t^\times v = v^\times t$,

$$\frac{\partial(R^\top b)}{\partial \Theta_k} = (t_k^\times)^\top R^\top b + R^\top \frac{\partial b}{\partial \Theta_k} \implies \boxed{D(R^\top b) = [(\bar{R}^\top \bar{b})^\times \bar{T}]_\Theta + \bar{R}^\top D b.} \quad (27)$$

9.1.2 Cable vector e_i

From $e_i = p_L + R_L \rho_i - p_i - R_i d_i$, applying (26) to each rotated constant:

$$\boxed{D e_i = [I]_{p_L} + [-\bar{R}_L \rho_i^\times \bar{T}_L]_{\Theta_L} + [-I]_{p_i} + [+ \bar{R}_i d_i^\times \bar{T}_i]_{\Theta_i}} \in \mathbb{R}^{3 \times 48}. \quad (28)$$

The sign flip on the last block comes from $e_i = c_i - h_i$: the hook offset enters with a minus sign, so its Jacobian enters with a plus.

9.1.3 Cable rate $\dot{e}_i$

Differentiating (17). The term $R_L(\omega_L \times \rho_i)$ is a rotation acting on the *state-dependent* body vector $a = \omega_L \times \rho_i = -\rho_i^\times \omega_L$, so both (26) and the product rule contribute:

$$D(R_L(\omega_L \times \rho_i)) = [-\bar{R}_L(\bar{\omega}_L \times \rho_i)^\times \bar{T}_L]_{\Theta_L} + [-\bar{R}_L \rho_i^\times]_{\omega_L}. \quad (29)$$

Doing the same for the drone side and subtracting:

$$\boxed{\begin{aligned} D\dot{e}_i = & [I]_{v_L} + [-\bar{R}_L(\bar{\omega}_L \times \rho_i)^\times \bar{T}_L]_{\Theta_L} + [-\bar{R}_L \rho_i^\times]_{\omega_L} \\ & + [-I]_{v_i} + [+ \bar{R}_i(\bar{\omega}_i \times d_i)^\times \bar{T}_i]_{\Theta_i} + [+ \bar{R}_i d_i^\times]_{\omega_i}. \end{aligned}} \quad (30)$$

Note that the Θ blocks of (30) are proportional to the body rates and therefore *vanish at any equilibrium*.

9.1.4 Length, direction and tension

Let $\bar{P}_i$ be the projector that differentiates a normalised vector,

$$\bar{P}_i = \frac{I - \bar{q}_i \bar{q}_i^\top}{\bar{l}_i} \quad \text{so that} \quad D\bar{l}_i = \bar{q}_i^\top D\bar{e}_i, \quad \boxed{Dq_i = \bar{P}_i D\bar{e}_i}. \quad (31)$$

$\bar{P}_i$ projects orthogonally to the cable and divides by its length, a change of $\bar{e}_i$ along the cable changes the length but not the direction. Differentiating $\tau_i = k(L_i - \ell_i) + c \dot{e}_i \cdot q_i$ with the product rule on the damping term:

$$\boxed{D\tau_i = k \bar{q}_i^\top D\bar{e}_i + c \left(\bar{q}_i^\top D\dot{e}_i + \bar{e}_i^\top \bar{P}_i D\bar{e}_i \right)} \in \mathbb{R}^{1 \times 48}. \quad (32)$$

The third term also vanishes at an equilibrium, where $\bar{e}_i = 0$.

9.1.5 Cable direction in the two body frames

Applying rule (27) with $b = q_i$, which is itself state-dependent:

$$\boxed{Ds_i = \bar{R}_i^\top Dq_i + [\bar{s}_i^\times \bar{T}_i]_{\Theta_i}, \quad Du_i = \bar{R}_L^\top Dq_i + [\bar{u}_i^\times \bar{T}_L]_{\Theta_L}.} \quad (33)$$

Where u_i is the cable direction for cable i in the load frame. Now all of the cable-state based derivatives are ready to be used to linearize the state equation and measurement equations.

9.2 Assembling the state matrix A

$A = \partial f / \partial x \in \mathbb{R}^{48 \times 48}$ is built one *row block* per dynamics equation. Each row block is read directly off Section 6.2 using the six cable Jacobians.

9.2.1 Payload translation

$\dot{p}_L = v_L$ is linear, giving $[I]_{v_L}$. For the velocity, differentiate (9); each cable contributes through both its tension magnitude and its direction:

$$D\dot{v}_L = -\frac{1}{m_L} \sum_{i=1}^3 \left[\bar{q}_i D\tau_i + \bar{\tau}_i Dq_i \right]. \quad (34)$$

The outer product $\bar{q}_i D\tau_i \in \mathbb{R}^{3 \times 48}$ is a rank-one term: a change in tension pushes the payload along the existing cable direction. The second term captures the cable direction rotating.

9.2.2 Payload attitude and rate

This is where the chart shows up. Differentiating $\dot{\Theta}_L = T_L^{-1} \omega_L$ requires differentiating a matrix inverse, $\partial(T^{-1}) = -T^{-1}(\partial T)T^{-1}$:

$$\boxed{D\dot{\Theta}_L = \left[\begin{array}{c|c} \cdots & -\bar{T}_L^{-1} \frac{\partial \bar{T}_L}{\partial \Theta_{L,k}} \bar{T}_L^{-1} \bar{\omega}_L \\ \hline \cdots \end{array} \right]_{k=1,2,3} \bigg|_{\Theta_L} + [\bar{T}_L^{-1}]_{\omega_L}. \quad (35)$$

The first block is a 3×3 matrix whose k -th column is the displayed vector. For the body rate, differentiating (11),

$$D\dot{\omega}_L = J_L^{-1} \left\{ -\sum_{i=1}^3 \left[(\rho_i^\times \bar{u}_i) D\tau_i + \bar{\tau}_i \rho_i^\times D u_i \right] + [(J_L \bar{\omega}_L)^\times - \bar{\omega}_L^\times J_L]_{\omega_L} \right\}. \quad (36)$$

The bracketed block is the linearised gyroscopic term, obtained from $\delta(-\omega \times J\omega) = [(J\bar{\omega})^\times - \bar{\omega}^\times J] \delta\omega$.

9.2.3 Drone translation

Differentiating (13), with $[I]_{v_i}$ for the position row:

$$\boxed{D\dot{v}_i = \frac{1}{m_i} \left\{ \bar{q}_i D\tau_i + \bar{\tau}_i Dq_i + [-\bar{R}_i F_i^\times \bar{T}_i]_{\Theta_i} \right\}. \quad (37)$$

The last term(rule (26) applied to the body thrust)is the thrust channel. Its null space is the thrust axis (body z), which is why this channel alone cannot see drone yaw.

9.2.4 Drone attitude and rate

Exactly parallel to the payload:

$$D\dot{\Theta}_i = \left[\begin{array}{c|c} -\bar{T}_i^{-1} \frac{\partial \bar{T}_i}{\partial \Theta_{i,k}} \bar{T}_i^{-1} \bar{\omega}_i \\ \hline \cdots \end{array} \right]_{\Theta_i} + [\bar{T}_i^{-1}]_{\omega_i}, \quad (38)$$

$$D\dot{\omega}_i = J_i^{-1} \left\{ (d_i^\times \bar{s}_i) D\tau_i + \bar{\tau}_i d_i^\times D s_i + [(J_i \bar{\omega}_i)^\times - \bar{\omega}_i^\times J_i]_{\omega_i} \right\}. \quad (39)$$

If the cable hook sits at the drone's centre of mass ($d_i = 0$) the cable exerts no torque and the drone attitude dynamics decouple entirely from the rest of the system.

9.3 Assembling the output matrix C

$C = \partial h / \partial x \in \mathbb{R}^{36 \times 48}$, one row block per sensor.

9.3.1 Motion capture on drone 1 (rows 1–6)

Position is read directly: $[I]_{p_1}$. By Rule 1 a coordinate change $\delta\Theta_1$ produces the rotation vector $\bar{T}_1\delta\Theta_1$,

$$\boxed{Dh_{\text{att}} = [\bar{T}_1]_{\Theta_1}} \quad (40)$$

9.3.2 UWB ranges (rows 7–9)

With antenna offset u_i^{ant} in the drone body frame, the antenna position and its Jacobian are

$$a_i = p_i + R_i u_i^{\text{ant}}, \quad Da_i = [I]_{p_i} + [-\bar{R}_i u_i^{\text{ant} \times} \bar{T}_i]_{\Theta_i}. \quad (41)$$

Differentiating $y_k = \|a_j - a_i\|$ for the pair $k = (i, j)$ gives a projection onto the line of sight,

$$\boxed{Dy_k = \hat{n}_k^\top (Da_j - Da_i), \quad \hat{n}_k = \frac{\bar{a}_j - \bar{a}_i}{\|\bar{a}_j - \bar{a}_i\|}.} \quad (42)$$

9.3.3 Cable directions, accelerometers, gyros (rows 10–36)

The body-frame cable direction is Ds_i from (33) directly. The accelerometer measures specific force, which for this model is $f_i^b = R_i^\top(\dot{v}_i - g) = (F_i + \tau_i s_i)/m_i$, so

$$Df_i^b = \frac{1}{m_i} [\bar{s}_i D\tau_i + \bar{\tau}_i Ds_i], \quad D\omega_i^{\text{meas}} = [I]_{\omega_i}. \quad (43)$$

10 The observability matrix

With A and C we can test the observability of the system with the observability matrix:

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{47} \end{bmatrix} \in \mathbb{R}^{1728 \times 48}, \quad \text{locally observable} \iff \text{rank } \mathcal{O} = 48. \quad (44)$$

Forming it numerically. Stacking CA^n naively is badly conditioned: the elastic cable modes make $\|A\| \sim 10^4$, so the later blocks overflow the earlier ones and the rank is underestimated. Two precautions suffice.

1. **Normalise** $A_n = A/\|A\|_2$. Scaling A by a positive constant is a time rescaling and does not change the observable subspace.

2. **Re-orthogonalise at every step** rather than stacking and factoring once:

$$Q \leftarrow \text{orth}(C), \quad Q \leftarrow \text{orth}([Q; QA_n]) \text{ until the rank stops growing.}$$

On this system the naive route reports 39 where the stable route correctly reports 40.

The Results Per Several Equilibrium Choices. The following table describes the rank of the observability matrix per several choices of equilibrium condition:

	Equilibrium	$\angle(F_i, q_i)$	rank
S1	vertical cables, equal ℓ_i	0°	40
S2	20° splay, equal ℓ_i	16.2°	48
S3	20° splay, unequal ℓ_i	16.2°	48
S4	generic point of the family	$16.9\text{--}18.1^\circ$	48
S5	vertical, equal ℓ_i , <i>with</i> offsets	0°	44
S6	20° splay, unequal ℓ_i , <i>with</i> offsets	16.2°	48

Table 1: Equilibrium evaluated versus resulting rank $\mathcal{O}$ (out of 48). $\angle(F_i, q_i)$ is the angle between each drone’s thrust axis and its cable, the scalar that predicts whether drone yaw is estimable. Rows S1–S4 carry no hardware offsets; “offsets” means the cable hook and UWB antenna are mounted off the drone’s **vertical axis**.

It is notable that the only two unobservable cases are when the cables are vertical, which is a condition that can be avoided by a controller and is already generally the condition of the CAMLS system.